RESEARCH ARTICLE

Figures and figure supplements

Taste quality and hunger interactions in a feeding sensorimotor circuit

Philip K Shiu *et al*

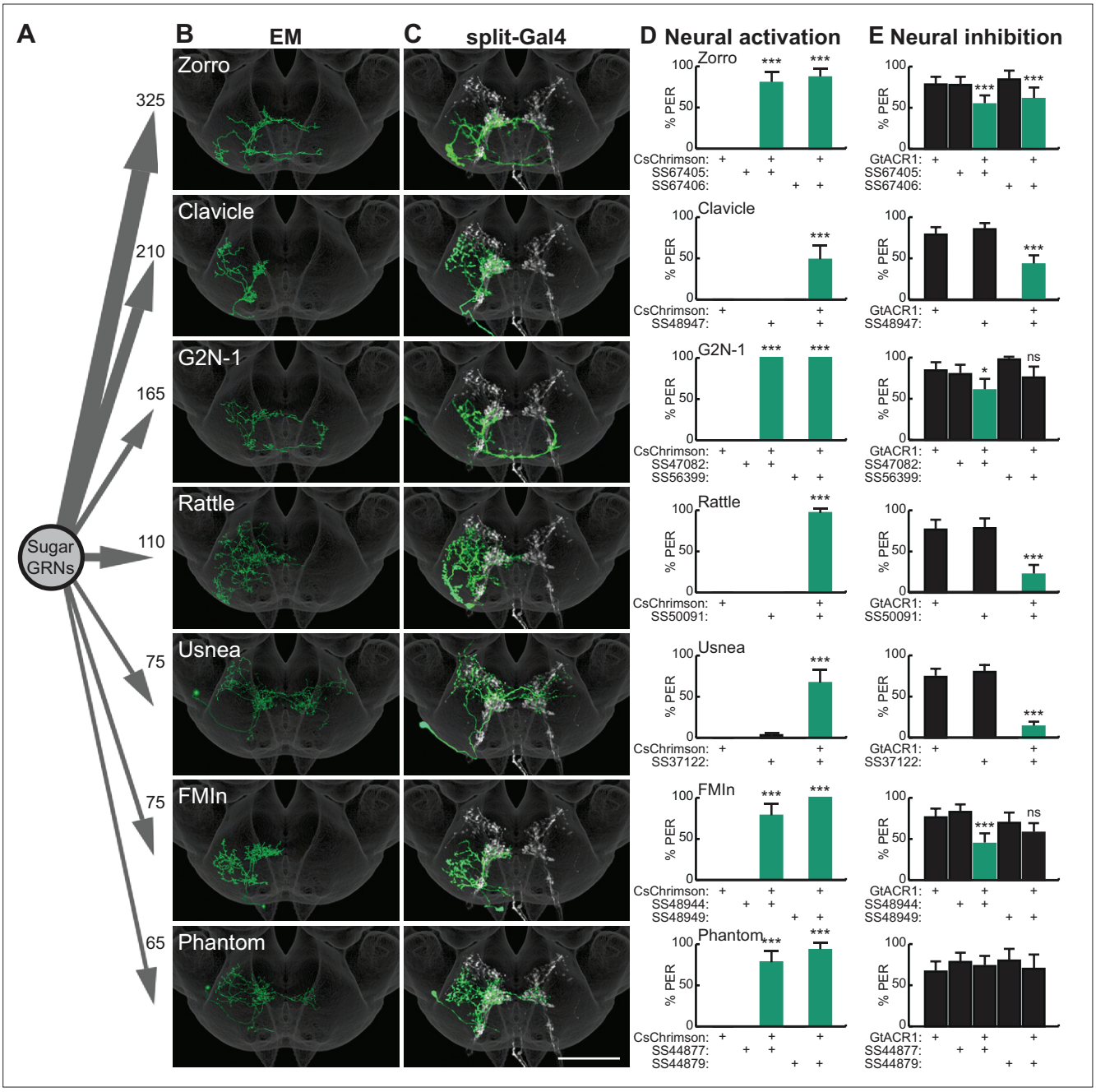


Figure 1. Sugar-sensing gustatory receptor neurons (GRNs) synapse onto multiple second-order neurons that influence proboscis extension. (**A**) Aggregate synaptic connectivity from sugar GRNs onto second-order sugar neurons. Numbers indicate the total number of synapses that the 17 candidate sugar GRNs make onto each second-order neuron. (**B–C**) Manually reconstructed electron microscopy (EM) skeletons (**B**) and registered neural images in split-Gal4 lines (**C**) for each second-order neuron in the subesophageal zone (SEZ) of the *Drosophila* brain. Sugar GRNs are depicted in white, JRC 2018 unisex coordinate space is shown in gray (**C**). Scale bar is 50 µm. (**D**) CsChrimson-mediated activation of seven second-order neurons elicits proboscis extension, n=30 flies per genotype. (**E**) GtACR1-mediated inhibition of second-order neurons reduces proboscis extension to 50 mM sucrose, n=46–83 flies per genotype. (**D–E**) The fraction of flies exhibiting proboscis extension response (PER) upon optogenetic or 50 mM sucrose stimulation. Mean ± 95% confidence interval (CI), Fisher's Exact Tests, *p<0.05, ***p<0.001. See ***Figure 1—figure supplement 1*** for EM reconstructions of additional second-order neurons and synaptic connectivity counts. See ***Figure 1—figure supplement 2*** for additional PER phenotypes of second-order sugar neurons.

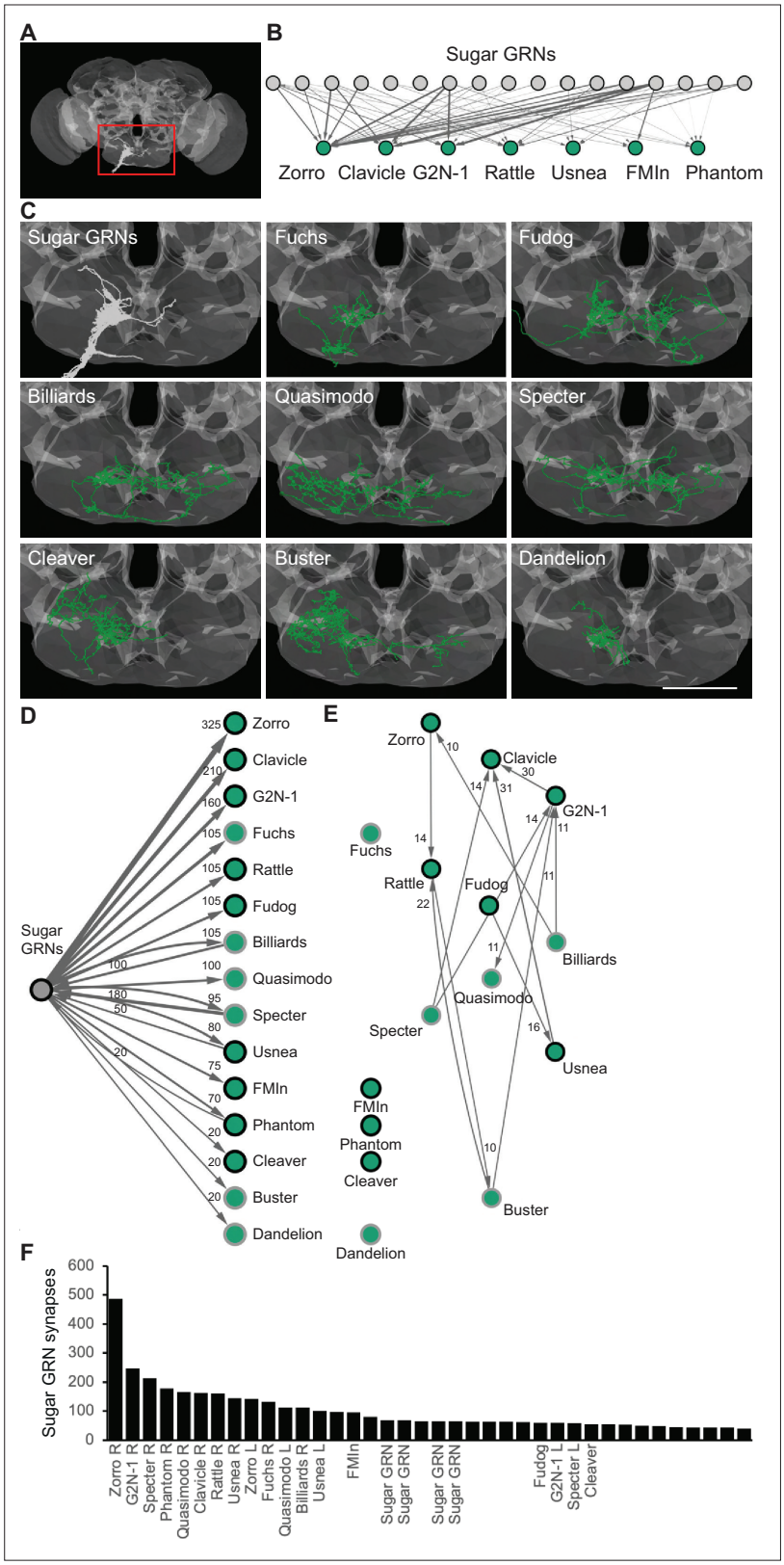


Figure 1—figure supplement 1. Anatomy of reconstructed second-order neurons and their connectivity, related to ***Figure 1***. (**A**) Schematic of the entire fly brain, showing electron microscopy (EM) reconstructed sugar-sensing gustatory receptor neurons (GRNs) in gray. Full adult fly brain (FAFB) neuropil space is shown in darker gray. Area outlined in red is enlarged in the first panel of C. (**B**) Synaptic connectivity of 17 previously identified candidate

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sugar GRNs onto second-order neurons that elicit proboscis extension response (PER). Line thickness represents the number of synapses, with a minimum of six synapses to a maximum of 46 synapses. (**C**) Anatomy of second-order candidate sugar EM reconstructed neurons. Scale bar is 50 µm. (**D**) Synaptic connectivity from sugar GRNs onto and from second-order neurons. Second-order neurons identified by EM and present in a split-Gal4 line (black circles); second-order neurons identified by EM only (gray circles). (**E**) Synaptic connectivity between second-order neurons. (**F**) Neurons with the most synapses from 17 candidate sugar GRNs based on Flywire predicted synapses (n≥40), with x-axis labeling neurons identified in this study.

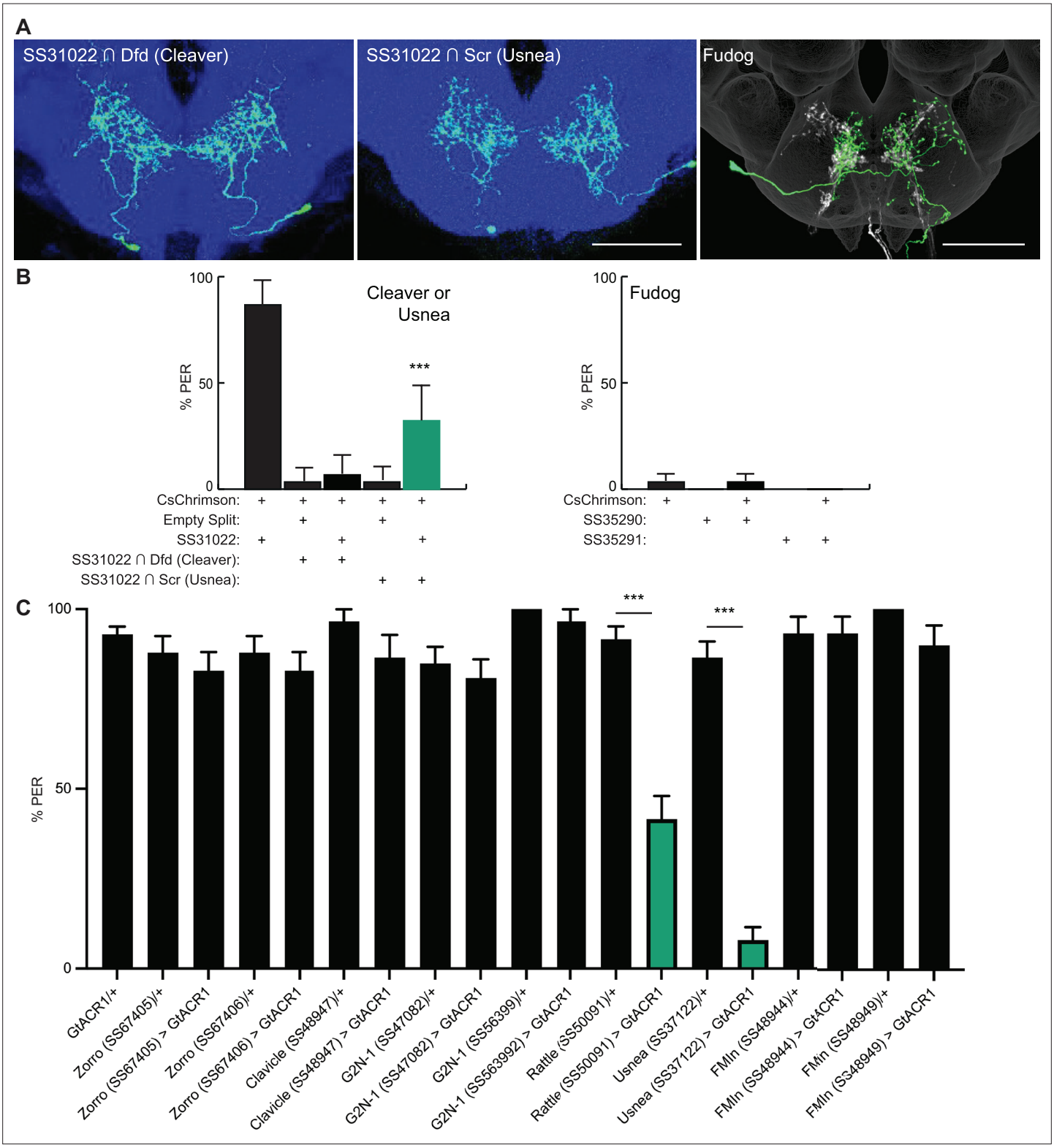


Figure 1—figure supplement 2. Additional proboscis extension phenotypes of second-order neurons, related to ***Figure 1***. (**A**) Light microscopy images of Cleaver, Usnea, and Fudog. Specific lines for Cleaver and Usnea were generated using a triple-intersection approach (see Methods). In the Fudog image, sugar GRNs are depicted in white. Scale bar is 50 µm. JRC 2018 unisex coordinate space is shown in blue (Cleaver and Usnea) or dark gray (Fudog). (**B**) Activation of Usnea, but not Cleaver or Fudog, elicits proboscis extension. n=30 flies per genotype. (**C**) Hyperpolarization of Rattle and Usnea inhibited proboscis extension to 100 mM sucrose, but hyperpolarization of other second-order neurons did not. n=30-144 flies per genotype. (B-C) Mean ± 95% confidence interval (CI), Fisher's Exact Tests, ***p<0.001.

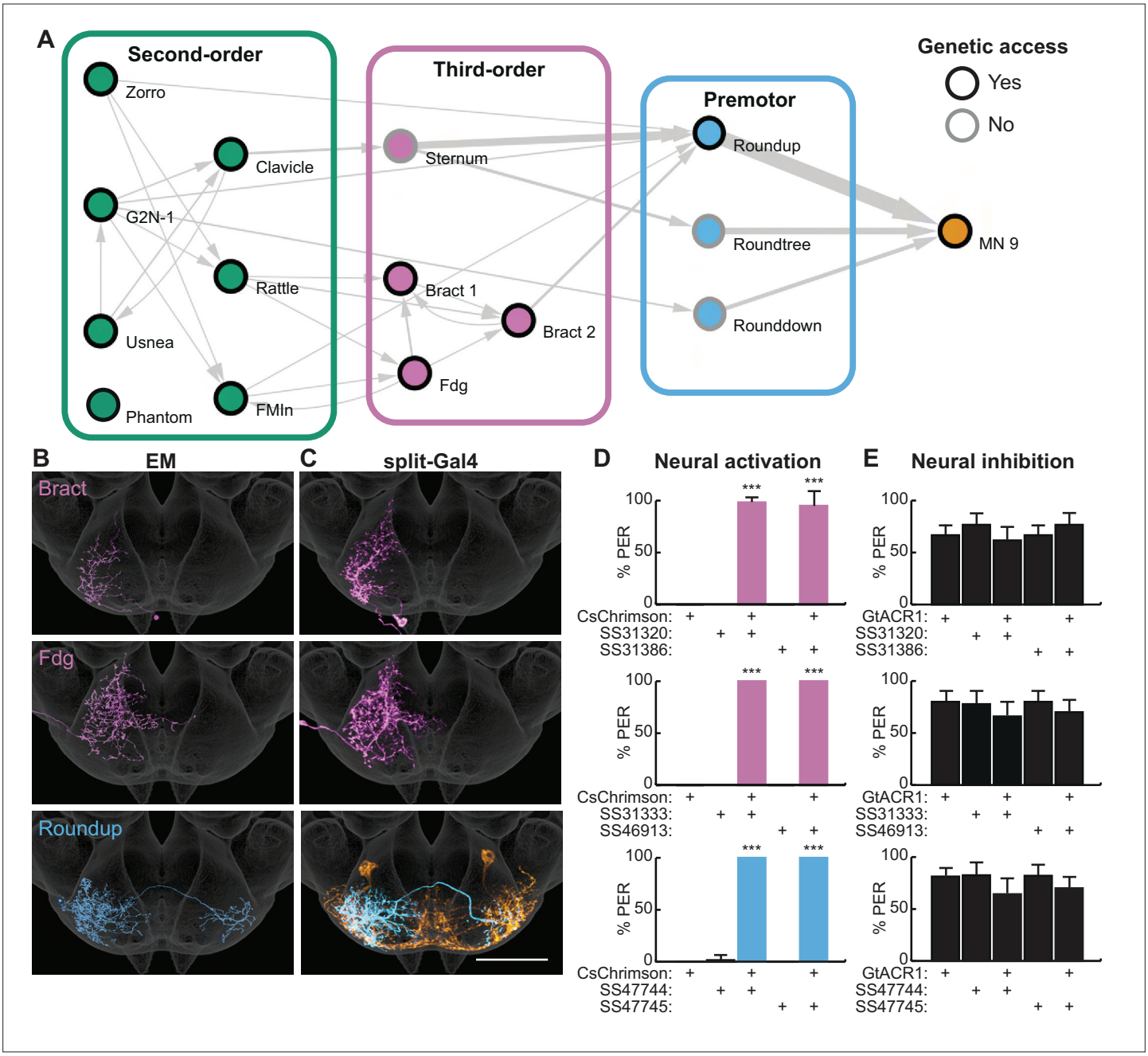


Figure 2. Second-order neurons synapse onto a local sensorimotor circuit for feeding initiation. (**A**) Schematic of the feeding initiation circuit. Circles outlined in black denote neurons with split-Gal4 genetic access, circles with gray outlines denote neurons without split-Gal4 genetic access. Line thickness represents synaptic connectivity of more than five synapses. (**B–C**) Electron microscopy (EM) neural reconstructions (**B**) and registered neural images in split-Gal4 lines (**C**) of third-order or premotor neurons in the subesophageal zone (SEZ). Scale bar is 50 µm. JRC 2018 unisex coordinate space is shown in gray, MN9 morphology is shown in orange. (**D**) CsChrimson-mediated activation of third-order or premotor neurons elicits proboscis extension response (PER), n=30 flies per genotype. (**E**) GtACR1-mediated inhibition of third-order or premotor neurons does not influence PER to 50 mM sucrose, n=40–70 flies per genotype. (**D–E**) Mean ± 95% CI, Fisher's Exact Tests, ***p<0.001. See ***Figure 2—figure supplement 1*** for synaptic counts.

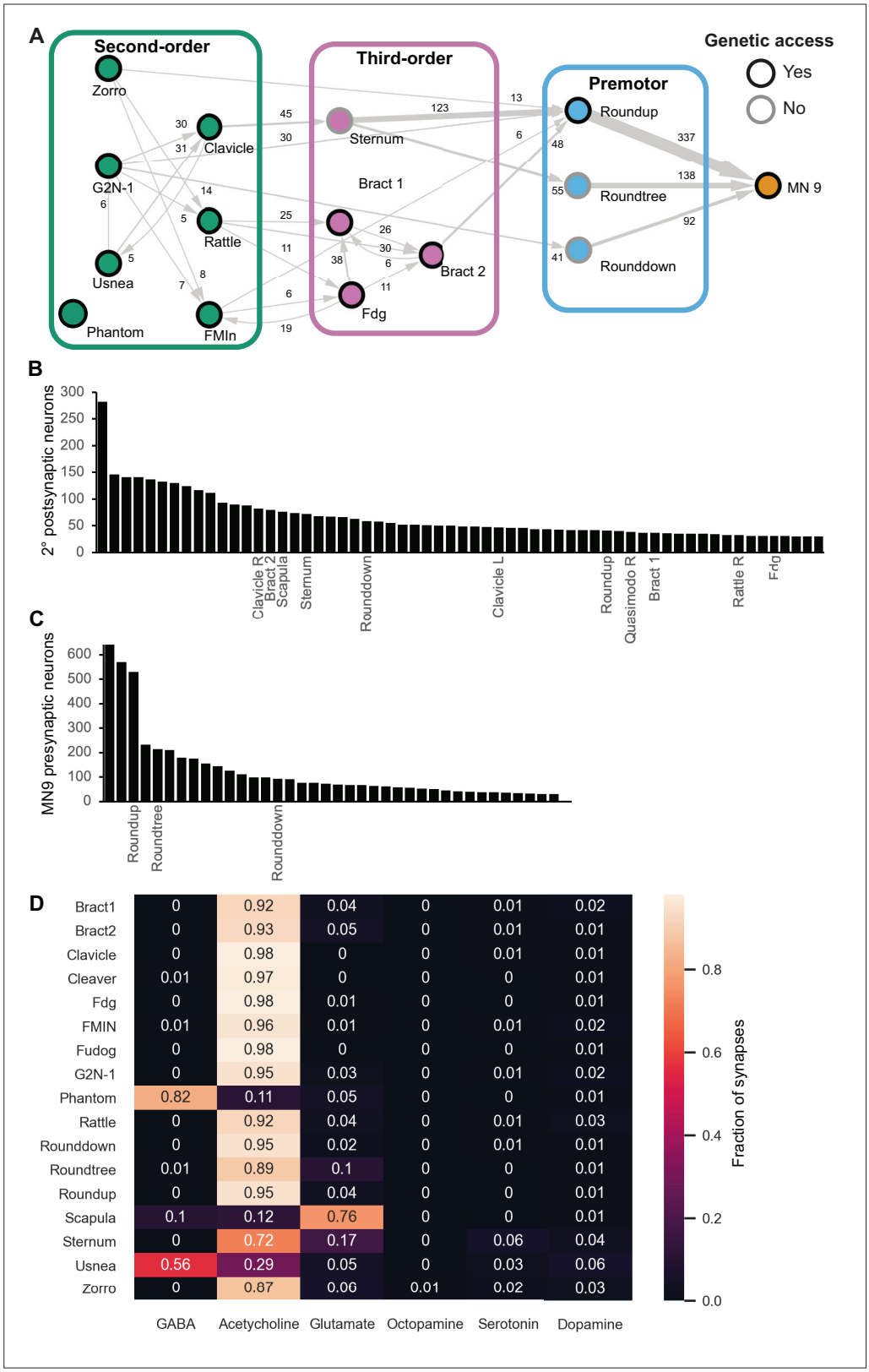


Figure 2—figure supplement 1. Synaptic connectivity in the feeding initiation circuit, related to ***Figure 2***. (**A**) Schematic of the feeding initiation circuit, with circles outlined in black for neurons with split-Gal4 lines, circles outlined in gray for neurons without split-Gal4 lines. Line thickness represents connectivity of more than five synapses, with synapse numbers labeled. (**B**) Neurons with the most synapses from second-order neurons that elicit

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proboscis extension response (PER), based on Flywire predicted synapses (n≥30), x-axis labels neurons identified in this study. (**C**) Neurons with the most synapses onto MN9, based on Flywire predicted synapses (n≥30), x-axis labels neurons identified in this study. (**D**) Neurotransmitter predictions (***Eckstein et al., 2020***) for individual synapses for each neuron in the feeding initiation circuit generated by a machine learning classifier. The fraction of synapses predicted to contain each neurotransmitter is indicated by color.

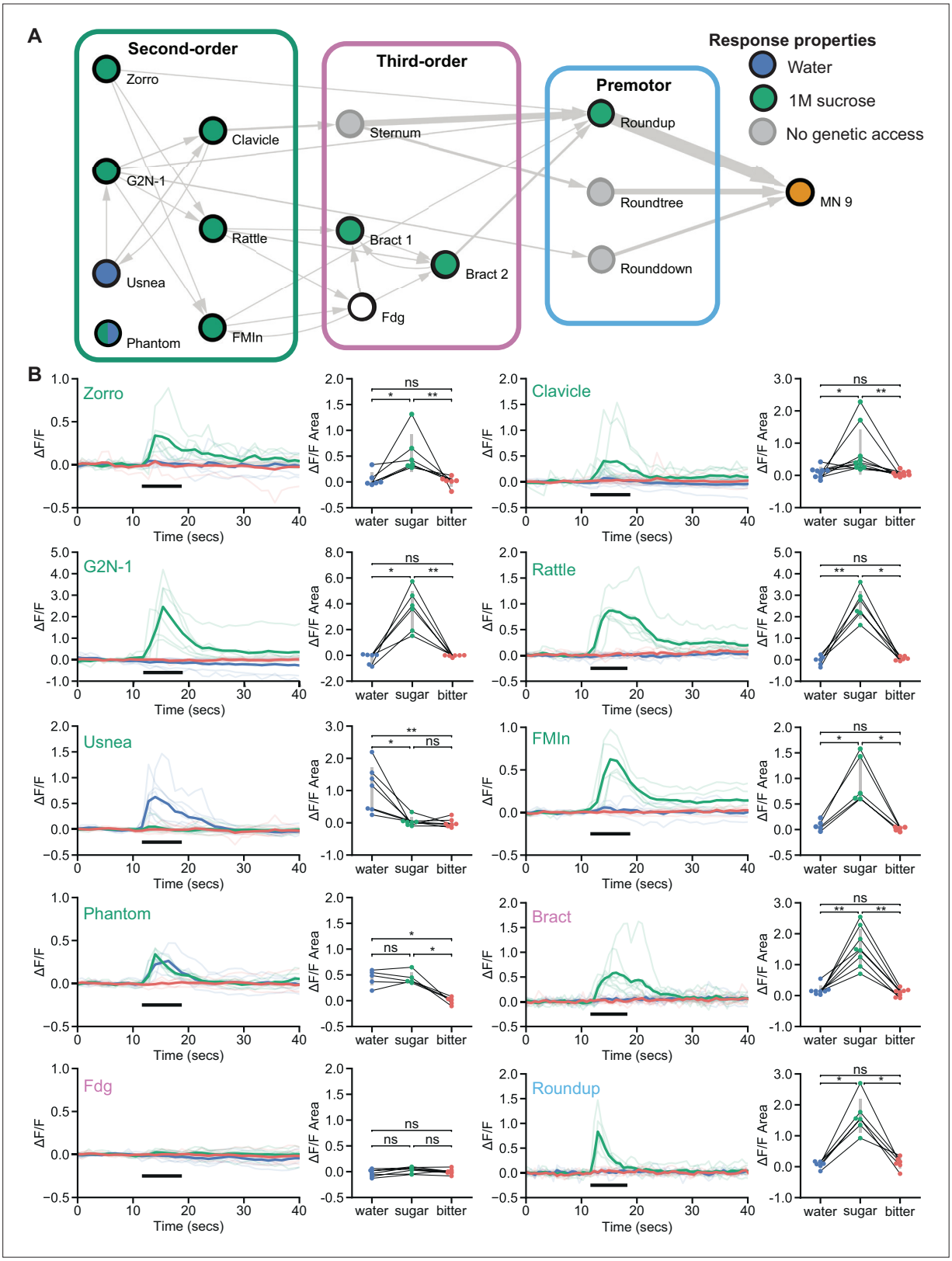


Figure 3. Feeding initiation neurons respond to taste detection. (**A**) Connectivity schematic of the feeding initiation circuit, where filled green circles represent cell types that respond to sugar detection, while filled blue circles represent cell types that respond to water detection. One cell type, Phantom, responds to both sugar and water (split blue and green circle). Fdg did not respond to proboscis taste detection (white circle), but see Figure S4A for responses to optogenetic activation of sugar gustatory receptor neurons (GRNs). (**B**) Calcium responses of feeding initiation neurons to

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stimulation of the proboscis in food-deprived flies. For each cell type, GCaMP6s fluorescence traces are shown on the left of the panel (ΔF/F), while ΔF/F area for each trace is shown on the right, with thin black lines indicating sample pairing. The proboscis of each tested individual was stimulated with water (green), sugar (blue), and bitter (red) tastants in sequential trials during the indicated period (thick black line). The following split-GAL4 lines were imaged for each cell type: Clavicle; SS48947, FMIn; SS48944, Zorro; SS67405, G2N-1; SS47082, Usnea; SS37122, Phantom; SS68204, Rattle; SS50091, Fdg; SS31345, Bract; SS31386, Roundup; SS47744. n=5-8 flies per genotype. Quade's test with Quade's All Pairs test, using Holm's correction to adjust for multiple comparisons, ns $p>0.05$, *$p\leq0.05$, **$p\leq0.01$. See also ***Figure 3—figure supplement 1*** for additional calcium imaging studies of feeding initiation neurons.

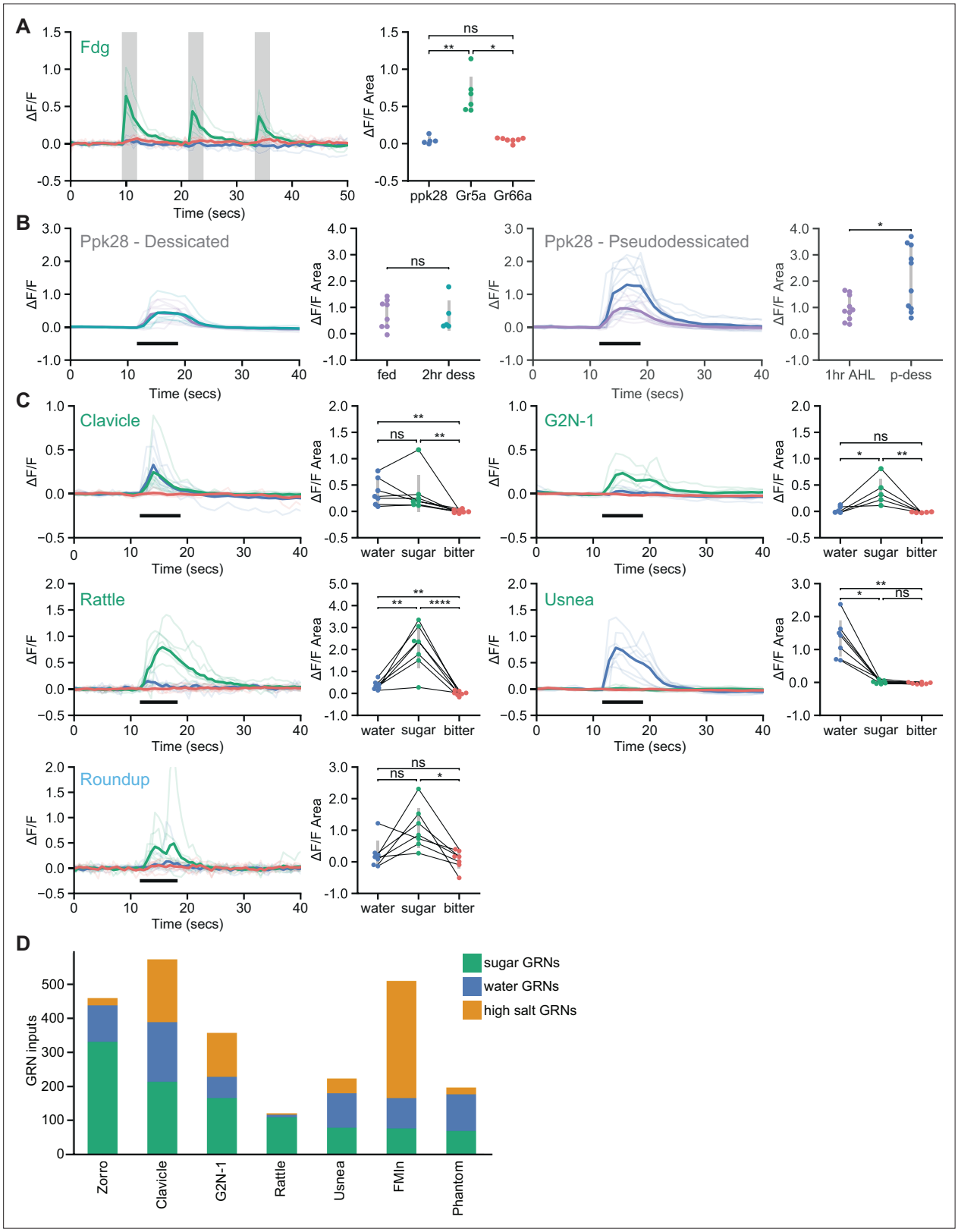


Figure 3—figure supplement 1. Taste responses of feeding initiation neurons, related to ***Figure 3***. (**A**) Calcium responses of Fdg to optogenetic activation of water (blue, ppk28-LexA), sugar (green, Gr5a-LexA), or bitter (red, Gr66a-LexA) GRNs in food-deprived flies. To examine Fdg responses, GCaMP7b was expressed using SS46913. Fluorescence traces are shown on the left of the panel (ΔF/F), while ΔF/F area for each trace is shown on the right. Stimulation with 660 nm light is indicated with vertical gray bars. Kruskal Wallace test with Dunn's test using Holm's correction to adjust for multiple

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comparisons, n=5-7 flies per genotype, ns p>0.05, *p≤0.05, **p≤0.01. (**B–C**) Calcium responses of water gustatory sensory neurons and feeding initiation neurons to taste stimulation of the proboscis. GCaMP6s fluorescence traces are shown on the left of each panel (ΔF/F), while ΔF/F area for each trace is shown on the right. n=5-10 flies per genotype, ns p>0.05, *p≤0.05, **p≤0.01, ***p≤0.001, ****p≤0.0001. (**B**) Taste responses to water in fed (purple) versus 2 hr desiccated (2 hr dess, teal) flies (left). Taste responses to water in fed flies allowed to rest for 1 hr in low osmolality artificial hemolymph after dissection before imaging (1 hr fed, purple) versus pseudodessicated, thirsty-like flies (p-dess, blue) (right). (**C**) Taste responses in pseudodessicated, thirsty-like flies. The proboscis of each tested individual was stimulated with water (green), sugar (blue), and bitter (red) tastants in sequential trials during the indicated period (thick black line). Thin black lines indicate sample pairing. The following split-GAL4 lines were imaged for each cell type: Clavicle; SS48947, G2N-1; SS47082, Usnea; SS37122, Rattle; SS50091, Roundup; SS47744. Quade's test with Quade's All Pairs test, using Holm's correction to adjust for multiple comparisons. (**D**) GRN synapes onto second-order neurons, with GRN categories based on ***Engert et al., 2021***.

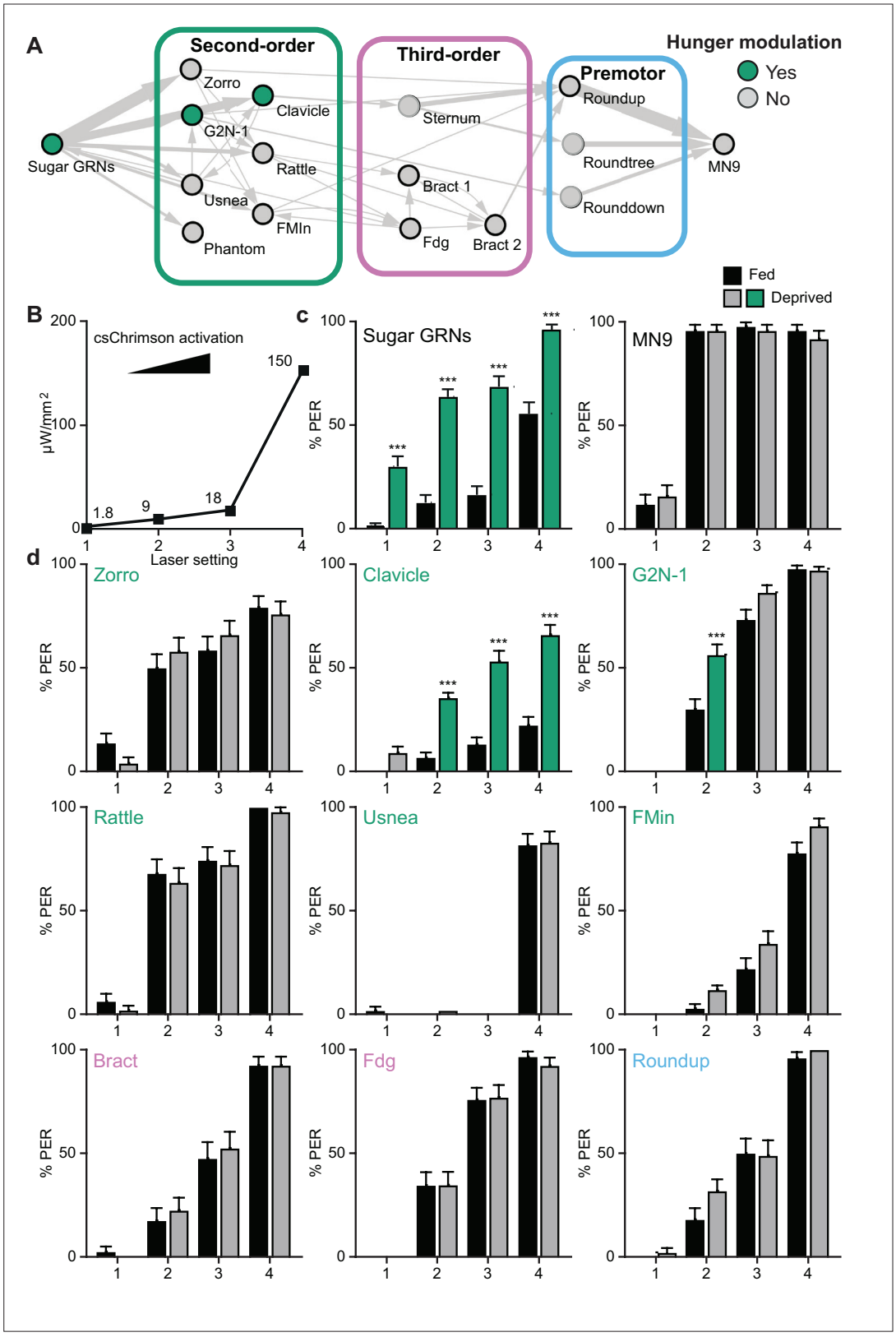


Figure 4. Hunger acts on a subset of second-order central neurons to modulate behavior. (**A**) Schematic of the feeding initiation circuit, with filled green circles representing nodes that are hunger-modulated. (**B**) Optogenetic activation at four different light intensities. (**C**) Activation of sugar-sensing neurons results in different feeding initiation rates between fed and food-deprived flies (left) whereas activation of MN9 does not (right), at four

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different light intensities. n=50. (**D**) Optogenetic activation of second-order, third-order, and premotor neurons in either fed or food-deprived flies. n=39–103. Mean ± 95% CI, Fisher's Exact Tests, ***p<0.001.

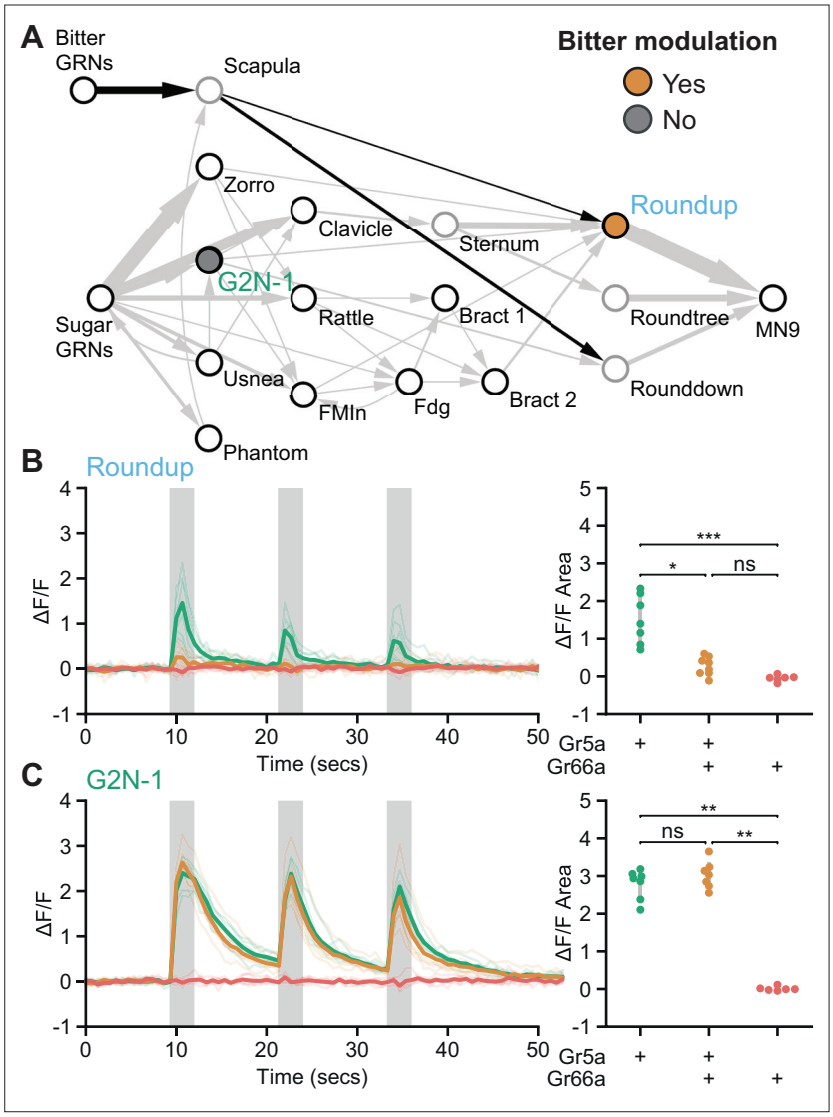


Figure 5. Premotor neurons integrate sweet and bitter taste detection. (**A**) Schematic of the feeding initiation circuit, showing a pathway from bitter GRNs to premotor neurons. Filled maize circle labels a premotor neuron inhibited by bitter tastants, filled gray circle labels an upstream second-order neuron that is not inhibited by bitter tastants. (**B and C**) Calcium responses of feeding circuit neurons to optogenetic activation of sugar (green, *Gr5a-LexA*), sugar plus bitter (maize, *Gr5a-LexA* plus *Gr66a-LexA*), or bitter (red, *Gr66a-LexA*) GRNs in food-deprived flies. For each cell type, Syt::GCaMP7b fluorescence traces are shown on the left of the panel (ΔF/F), while ΔF/F area for each trace is shown on the right. Periods of stimulation with 660 nm light are indicated with vertical gray bars. (**B**) SS47744 was imaged to examine Roundup responses. (**C**) SS47082 was imaged to examine G2N-1 responses. (B-C) Kruskal Wallace test with Dunn's test using Holm's correction to adjust for multiple comparisons, n=6-8 flies per genotype, ns $p>0.05$, *$p\leq0.05$, **$p\leq0.01$, ***$p\leq0.001$. See ***Figure 5—figure supplement 1*** for synaptic counts of second-order bitter neurons.

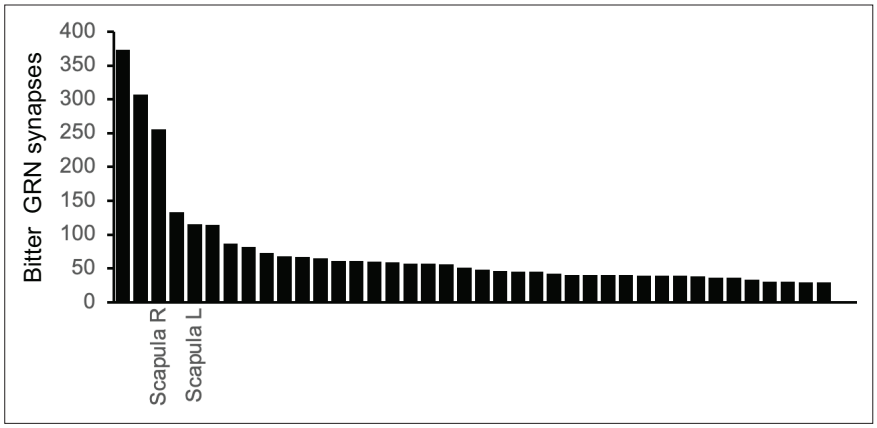


Figure 5—figure supplement 1. Second-order bitter neurons, related to ***Figure 5***. Neurons with the most synapses from candidate bitter gustatory receptor neurons (GRNs) based on Flywire predicted synapses (n≥30), with x-axis labeling neurons identified in this study.